



EPA Public Access

Author manuscript

J Contam Hydrol. Author manuscript; available in PMC 2025 January 01.

About author manuscripts

Submit a manuscript

Published in final edited form as:

J Contam Hydrol. 2024 January ; 260: 104268. doi:10.1016/j.jconhyd.2023.104268.

Geophysical Signatures of Soil AFFF Contamination from Spectral Induced Polarization and Low Field Nuclear Magnetic Resonance Methods

Samuel Falzone^{1,*}, Charles Schaefer², Ethan Siegenthaler¹, Kristina Keating², Dale Werkema³, Lee D. Slater^{1,4,*}

¹Department of Earth and Environmental Sciences, Rutgers University Newark, Newark, NJ

²CDM Smith, Edison, NJ.

³Environmental Protection Agency (EPA), Newport, OR 97365

⁴Pacific Northwest National Laboratory, 902 Battelle Blvd, Richland, WA 99354, USA

Abstract

Few field methods are available for characterizing source zones impacted with aqueous film forming foam (AFFF). Non-invasive geophysical characterization of AFFF source zone contamination in situ could assist with the delineation and characterization of these sites, allowing for more informed sampling regimes aimed at quantifying subsurface poly- and perfluoroalkyl substances (PFAS) contamination. We present initial results from the investigation of the sensitivity of two existing surface and borehole-deployable geophysical technologies, spectral induced polarization (SIP), and low field nuclear magnetic resonance (NMR), to soils impacted with AFFF. To investigate the sensitivity of these methods to AFFF-impacted soil, bench-scale column experiments were conducted on samples consisting of natural and synthetic soils and groundwater. While our findings do not show strong evidence of NMR sensitivity to soil PFAS contamination, we do find evidence that SIP has sufficient sensitivity to detect sorption of AFFF constituents (including PFAS) to soils. This finding is based on evidence that AFFF constituents associated with the pore surface produce a measurable polarization response in both freshly impacted synthetic soils and in soils historically impacted with AFFF. Our findings encourage further exploration of the SIP method as a technology for characterizing contaminant concentrations across AFFF source zones.

1. Introduction

1.1. Geophysical characterization of PFAS source zones

The release of aqueous film forming foam (AFFF) into the environment is an issue for water resources management that is continuing to grow in importance in the United States

*Corresponding author: Samuel Falzone; s.o.falzone@gmail.com, Lee D. Slater; lslater@newark.rutgers.edu.

Supporting Information

Detailed description of column construction (Text S1) and geophysical measurement procedures (Text S2); site location for the samples acquired from a PFAS source zone (Figure S1); table of total PFAS concentrations recorded for six soil samples and two groundwater samples (Table S1).

and worldwide. In particular, the use of AFFF as a fire suppressant has a high potential for contaminating water resources (Houtz et al., 2013). The poly- and per- fluoroalkyl substances (PFASs) associated with AFFF releases persist in the soil and serve as a long-term source of contamination to underlying groundwater (Anderson et al., 2016). PFAS are long lasting chemicals which break down slowly over time and are found in the blood of people and animals worldwide, as well as in water, fish and soils (Houtz et al., 2013; Taylor, 2020; Death et al., 2021; Schaefer et al., 2022). The exposure to PFAS in the environment is increasingly linked to harmful health effects in humans and animals (Death et al., 2021).

PFAS associated with AFFF contamination have been shown to persist in soils (Anderson et al., 2016; Barzen-Hanson et al., 2017; Brusseau et al., 2019) where association with the soil mineral surface serves as an important factor controlling their fate and transport (Backe et al., 2013; Houtz et al., 2013; Tang et al., 2010). Despite the importance of PFAS contamination in the environment as it pertains to water resource management, the nature and distribution of PFAS in AFFF-impacted source zones remains poorly understood. This lack of understanding is due in large part to the absence of effective field-scale AFFF source zone characterization methods above and beyond blind direct sampling and subsequent laboratory analysis (Day-Lewis et al., 2017).

Geophysical technologies can contribute to environmental site characterization by non-invasively providing information on the variation in subsurface physical properties (e.g., electrical resistivity) that can be used to guide sampling activities towards likely contamination hotspots. Our work is motivated by the possibility that variations in AFFF contamination across source zones might be detectable from geophysical measurements. As far as we are aware, no previous studies have investigated the sensitivity of geophysical measurements to AFFF contamination. We recognize that geophysical techniques are not sufficiently sensitive to PFAS, even at the relatively high concentrations observed in soils, to be considered comparable to analytical methods for PFAS quantification with associated sensitivity and detection limits. However, the net impacts of soil contamination with AFFF, considering all its constituents, may sufficiently alter the physicochemical properties of the soils to be detectable with geophysics.

Two geophysical technologies, low field nuclear magnetic resonance (NMR) and spectral induced polarization (SIP), are sensitive to the physicochemical properties of the soil mineral surface, including changes due to contaminant interactions that is of particular interest for AFFF characterization. Both SIP and NMR have proven sensitivity to surface chemical and mineralogical properties of pore surfaces (Keating and Knight, 2007; Zhang et al., 2012), and have demonstrated success for monitoring sorption processes in situ (Ben Moshe and Furman, 2022; Bryar and Knight, 2008; Hao et al., 2015; Kirmizakis et al., 2019; Mellage et al., 2022, 2018; Vaudelet et al., 2011). Consequently, these methods may have sensitivity to sorption of PFAS and/or sorption of the persistent non-fluorinated hydrocarbon surfactants (HCS) associated with historically impacted AFFF impacted source areas (García et al., 2019). Many AFFF impacted source zones have been shown to contain elevated concentrations of PFAS, raising the possibility that geophysical detection of PFAS itself may be possible. For example, within a source zone at Ellsworth Air Force Base, the sum of all PFAS in soils is greater than 70,000 µg/kg (McGuire et al., 2014), potentially

bringing the task of characterizing the extent of soil PFAS contamination within the domain of indirect geophysical sensing. However, we consider the combination of elevated PFAS levels with elevated levels of co-located HCSs to be the probable cause of any SIP and/or NMR signals detectable within AFFF source areas.

2. Theory and Calculations

2.1 Spectral Induced Polarization

The SIP technique is unique in applied geophysics in that it has strong sensitivity to both the geometry of the pore space and the electrochemistry of the mineral-fluid interface. Consequently, there has been interest in using SIP to investigate geochemical transformations, due in large part to the demonstrated sensitivity of the method to sorption processes (Ben Moshe and Furman, 2022; Hao et al., 2016, 2015; Kirmizakis et al., 2019; Mellage et al., 2022, 2018). For example, research on the SIP method has demonstrated its potential for studying geochemical and biogeochemical processes that alter both physical and chemical properties of the mineral-fluid interface (Atekwana and Slater, 2009; Kessouri et al., 2019).

The SIP is a four-electrode technique that records the complex impedance of a sample (or the subsurface) over a range of frequencies (typically less than 1000 Hz). The complex conductivity (σ^*) of the sample is computed from the impedance, which contains real (σ') and imaginary (σ'') conductivity terms,

$$\sigma^* = \sigma' + i\sigma'' . \quad (1)$$

where the real term accounts for conduction processes and the imaginary (i) term accounts for temporary charge storage (polarization) processes in a soil. Below 1000 Hz, conduction strongly dominates charge transport in soils, requiring accurate measurements of small phase shifts to reliably determine σ'' .

Models of charge transport in soils include both electrolytic (σ_{el}) and complex surface (σ_{surf}) conductivity terms, associated with the electrical properties of the pore fluid and ions within the electrical double layer (EDL) respectively. The imaginary component of σ_{surf} is proportional to the surface-area-to-pore volume ratio (S_{por}) divided by the formation factor (F), with a proportionality constant known as the specific polarizability (c_p) (Weller and Slater, 2019),

$$\sigma''_{surf} = c_p S_{por} / F . \quad (2)$$

c_p depends upon the electrochemical properties of the EDL such as the surface charge density and mobility, accounting for the sensitivity of SIP to sorption processes.

2.2 Low Field Nuclear Magnetic Resonance (NMR)

Low field NMR geophysical field methods are directly sensitive to the hydrogen ions present in water. Unlike NMR spectroscopy, which relies on the use of strong magnetic field gradients to measure in the frequency domain with the intent of determining the molecular identity and structure of isotopes, geophysical NMR measurements use the principles of NMR relaxometry. This method involves analysis of the exponentially decaying signal measured in the time domain and characterized by the relaxation time (T_2); the decay results when energized hydrogen atoms realign with a static magnetic field following the application of a series of oscillating secondary magnetic field pulses²². Inverting the signal measured from water saturated porous media typically results in a T_2 -distribution, which is related to the distribution of physicochemical pore environments (pore size and surface mineralogy) within the sample volume. The mean-log relaxation time of the T_2 -distribution (T_{2ML}) is thus related to the mean physical and surface geochemical characteristics of the pore space. T_{2ML} is inversely proportional to the surface-area-to-volume ratio (S_{por}) and the surface relaxivity (ρ_2), which defines the ability of the pore surface to enhance relaxation through the magnetic coupling of protons and paramagnetic minerals on the pore surface (Kleinberg and Horsfield, 1990),

$$T_{2ML}^{-1} \approx S_{por}\rho_2. \quad (3)$$

In most cases, sorption of paramagnetic particles to the surface has been shown to increase ρ_2 and augment T_2 (Keating and Knight, 2007). However, the sorption of contaminants has had a mediating effect on relaxation by reducing ρ_2 , creating longer observed T_2 (Bryar and Knight, 2008). This phenomenon has been attributed to the ability of surface coatings of contaminants to inhibit proton/paramagnetic coupling. Thus, there is a potential for surficial transformations associated with sorbed anthropogenic contamination, such as PFAS/HCS, to produce a measurable low field NMR response.

2.3 Sensitivity to Sorbed AFFF Constituents

Variations in c_p (Eq. 2) known to result from sorption processes and variations in ρ_2 (E1. 3) could conceivably be influenced by sorption of certain contaminants. Thus, SIP and NMR measurements may sense ionic PFAS/HCS sorption onto soil surfaces if this results in modifications to the surface chemistry that are manifested as a measurable change in c_p and/or ρ_2 . Fig. 1 illustrates the conceptual responses of SIP (a) and NMR (b) measurements resulting from the presence of soil-sorbed AFFF contaminants. The hypothesized polarization enhancement observed with SIP is attributed to a larger net displacement of charge within the EDL in response to the applied electric field (E_0) as a result of ionic PFAS/HCS sorption. The resulting increase in c_p would result in a positive correlation between σ'' and sorbed ionic PFAS or HCS concentration. The hypothetical mechanism of NMR measurements to sorbed PFAS/HCS is less direct than SIP, relying on the ability of sorbed ionic PFAS/HCS to alter ρ_2 by inhibiting proton/paramagnetic coupling.

Under these circumstances, a positive response may appear as a shift in T_2 from shorter to longer times associated with diminished ρ_2 .

In this study, we present a series of bench-scale column experiments on synthetic and natural soils to evaluate the potential for using SIP and/or NMR for detecting the presence of soil AFFF contamination. Here, we explore the geophysical response associated with the level of contamination characteristic of a fresh AFFF impacted site, as well as contamination associated with a source zone at an AFFF impacted site.

3. Material and methods

3.1 Soil and groundwater samples

A series of column experiments were performed in triplicate to assess the feasibility of using SIP and/or NMR for characterizing AFFF source zones. These experiments were conducted on both synthetic soils and natural soils sampled from an AFFF source zone location. Soils were saturated with either AFFF impacted groundwater, or with synthetic groundwater based on the ionic concentrations of groundwater sampled from two field sites. Each column experiment was designed to determine the sensitivity of SIP and NMR to the presence of AFFF contamination under conditions associated with either (1) fresh AFFF release, or (2) the source zone of an AFFF impacted site.

The synthetic soil was a well-characterized sand mixed with organic matter (Premier Tech Horticulture, USA) (10% concentration by volume), to increase surface area and promote sorption. Wedron silica sand (Wedron Silica Co.), a medium, well-sorted, round, clean sand with a narrow grain-size distribution centered at 300 μm was used. The sand and organic matter in the synthetic soils were mixed by hand and allowed to dry at room temperature prior to saturation.

Six natural soils were collected from a site located in the northeastern United States historically impacted with AFFF, but where no known firefighting activities occurred. Soils were collected approximately 25 years after the last known AFFF applications (Fig. S1 shows sample locations). Sampling of the site was designed for collection of both “control” and “experimental” samples. The experimental group consists of three samples (I1, I2, I3) collected from within the confirmed source zone, while the control group consists of three additional samples (C1, C2, C3) collected from the same site, but at a location outside of the source zone. All samples were collected from the same soil horizon at a depth of 0.3 m. Laser diffraction particle size analysis confirmed that the physical characteristics of all soils is similar, with mean grain diameters of $426 \pm 50.7 \mu\text{m}$. Total PFAS analysis was performed on leachates from all soils by SGS AXYS (Sidney, Canada) using Environmental Protection Agency (EPA) Method 537 (EPA, 2022). Impacted sample total PFAS concentrations varied from 72–1340 $\mu\text{g/kg}$ whereas control samples varied from 1.37–5.21 $\mu\text{g/kg}$ (Table S1). PFAS constituent concentrations for soil samples are provided in Table S2.

Saturating fluids included AFFF impacted site groundwater, synthetic groundwater, and an AFFF-impacted synthetic groundwater. Synthetic groundwaters were based on AFFF impacted groundwater sampled from two field sites (i.e. GW1 and GW2). Groundwater

GW1 was sampled from a well screened at approximately 2.4 m below ground surface (bgs) in the source zone of the AFFF impacted site from where AFFF-impacted soil samples were acquired. Groundwater GW2 was sampled from a shallow groundwater well screened at approximately 5.5 m bgs in the source zone of another historically AFFF-impacted site. Major cation constituents of the groundwater samples and their concentrations were determined spectrophotometrically using inductively coupled plasma mass spectrometry (ICP-MS). Total PFAS measured for each groundwater sample is reported in Table S1, while individual concentrations of dominant PFAS constituents are reported in Table S2 (again performed by SGS-AXYS using EPA Method 537).

Synthetic (unimpacted) groundwater solutions were prepared based on chemical analysis of groundwater GW1 and GW2. Synthetic (unimpacted) groundwater based on GW1 was used for column experiments on synthetic soils saturated with AFFF impacted groundwater, and for saturating the field samples. GW2 and synthetic (unimpacted) groundwater based on GW2 were used for column experiments on synthetic soils saturated with directly sampled AFFF impacted groundwater. The ionic composition of GW2 is reported in Table S3 (the ionic composition of GW1, while serving as the basis for a synthetic groundwater, is omitted as it was not directly used as the saturating fluid in any experiment).

Synthetic groundwaters were prepared in bulk by mixing major cationic constituents based on spectrophotometric laboratory analysis of both groundwater samples (Table S4). The fluid specific conductance of GW1 and GW2 (1.41 and 1.37 mS/cm respectively) was used to verify similarities in the ionic concentrations of the sampled groundwaters and their synthetic analogs. A sub-volume of synthetic groundwater based on GW1 was contaminated with a 500-fold dilution of a previously characterized AFFF solution containing several polyfluorinated zwitterionic and cationic PFAS (Table S5)(Schaefer et al., 2018). A 500-fold dilution was selected to attain PFAS concentrations representative of groundwater sampled from an AFFF source zone, such as ~11 mg/L total perfluoroalkyl acids (PFAAs), as confirmed in previous studies from impacted sites (Backe et al., 2013). The introduction of AFFF to the synthetic groundwater raised its specific conductance slightly from 1.37 mS/cm to 1.43 mS/cm.

3.2 Column Experiments

3.2.1 Column design and sample packing—Sample columns were designed to accommodate both low field nuclear magnetic resonance (NMR) and spectral induced polarization (SIP) laboratory measurements. Columns were milled from polycarbonate, with an inner diameter of 2.4 cm and an internal height of 10 cm, resulting in an internal volume of 45.2 mL (Figure S2). Each column included four embedded electrodes: two permanent current electrodes positioned individually at the top and bottom of the column and two removable potential electrodes positioned near the center of the column, spaced 2 cm apart. Electrodes were constructed from 3.2 mm diameter silver chloride (AgCl) coated Ag wire, prepared by coating 99.9% pure Ag wire in bleach for 8 hours. Soils were wet packed into columns by mixing each soil in a basin with the desired saturating fluid and then tamping the slurry into the columns in 2 cm long segments. All samples were prepared in triplicate for each experimental group. Porosity of each packed sample was determined

gravimetrically using the wet and dry weights prior to and following column preparation. Columns within each sample set were confirmed to have similar porosities (i.e., $40.8 \pm 2.9\%$ for natural soils and $53.7 \pm 1.9\%$ for synthetic soils) after preparation in order to minimize error associated with sample preparation.

3.2.2. AFFF Impacted Groundwater Experiment—Six columns were prepared using the synthetic soil composed of sand and organic soil. The columns were divided into two groups of three to give triplicate measurements. The experimental group consisted of three columns saturated with the AFFF impacted synthetic groundwater based on groundwater GW1. As a control to this group, three additional columns were saturated with clean synthetic groundwater, GW1. The SIP and NMR measurements of the columns packed with synthetic soils were periodically collected over 12 days to allow time for full chemical equilibrium between the saturated fluid and the soil particle surface to be reached. The periodic measurements were intended to capture any evolution of the geophysical signatures as the saturating solution chemistry comes into equilibrium with the surface.

3.2.3. AFFF Impacted Site Contaminated Groundwater Experiment—A similar column experiment to the AFFF impacted groundwater experiment was conducted using groundwater GW2. The experimental group consisted of a subset of three columns saturated using the AFFF impacted site groundwater GW2. As a control to this group, three additional columns were saturated with a synthetic groundwater created based on GW2. As was the case for the AFFF impacted groundwater experiment, SIP measurements of the columns were periodically acquired, over 10 days in this case, to approach surface equilibrium.

3.2.4 Analysis of AFFF Impacted Samples—Triplicate columns of each of the six natural soils collected from the AFFF impacted site (Fig. S1) were analyzed with SIP and NMR measurements. These columns are divided into two groups: the experimental group consisting of the source zone samples (I1, I2, I3) and the control group consisting of samples collected from outside the source zone (C1, C2, C3). The clean synthetic groundwater based on GW1 was used as the saturating fluid for all columns. All PFAS/HCS present in the soils is therefore exclusively the result of the AFFF impacted site contamination associated with the source zone. The SIP and NMR measurements were collected immediately after preparation to minimize the effects of any PFAS/HCS desorption. In contrast to the AFFF impacted groundwater experiment, SIP and NMR measurements were not repeated over time, as PFAS/HCS was already in equilibrium and sorbed to the soil surface.

3.3 Data Acquisition and Analysis

3.3.1 Spectral Induced Polarization—Spectral induced polarization measurements were acquired with an instrument manufactured by Ontash & Ermac Inc. (NJ, USA). Measurements of the impedance magnitude and phase shift were acquired from 0.01 Hz to 10,000 Hz using a four-electrode configuration. Alternating current was injected from the top and bottom electrode of each column using an applied voltage of 5 V, while the resulting voltage difference was measured between the potential electrodes. The σ^* was determined using the measured impedance magnitude, and phase shift, along with the geometric factor computed for the column based on 1D current flow.

3.3.2 Nuclear Magnetic Resonance—Nuclear magnetic resonance measurements were acquired with a Magritek Rock Core Analyzer, using the CPMG pulse sequence. Each echo was sampled using 32 data points and an echo time of 200 μ s. An inter-stack delay time of 10 s was used to assure equilibrium alignment to the B_0 field prior to each stack. In order to fully sample the decay curves, measurements collected on the synthetic experiments used 5,000 echoes, while measurements collected on natural soils used 20,000 echoes. All measurements were collected using 16 stacks. A non-negative least squares inversion algorithm with second order Tikhonov regularization was used to fit the data with 160 log-spaced values of T_{2i} ranging from 10^{-4} to 1 s. The regularization parameter, which determines the degree of smoothness, was chosen using the L-curve corner criteria such that both the model norm and the residual norm were minimized. T_{2ML} was calculated from the T_2 -distributions resulting from inversion.

4. Results and Discussion

4.1 Synthetic samples

4.1.1 AFFF Impacted Soils—The SIP results on columns packed with the synthetic soil are shown for the 12-day period following preparation in Fig. 2. The colors represent the AFFF impacted (pink) and control (black) samples and the error bars show the standard deviation of the measurements on triplicate samples. The samples saturated with AFFF impacted synthetic groundwater based on GW1 (Fig. 2b) exhibit a stronger polarization (higher σ''), and also show different spectral characteristics than the columns saturated with clean synthetic groundwater (Fig. 2a). A stronger polarization peak (around ~ 20 Hz) is evident in the AFFF impacted soils. This same peak is less evident in the control group columns, saturated with clean synthetic groundwater GW1. Error analysis indicates that the variance between triplicate columns is low (the exception being the day 12 measurement from the control group). Increases in σ' are also observed for the AFFF impacted soils, although the relative increases are substantially less than those observed for σ'' .

Given that previous studies strongly link σ'' to sorption (Hao et al., 2016), coupled with previous works describing sorption as the dominant mechanism of long-term soil-PFAS contamination (Campos Pereira et al., 2018), the differences in σ'' between impacted and unimpacted columns likely indicate that σ'' is sufficiently sensitive to the sorption of AFFF constituents not present in the unimpacted groundwater. However, it is unclear whether the polarization enhancement observed in Figure 2b results from sorbed ionic PFAS constituents and/or sorption of ionic HCS also present in AFFF. The measured response is consistent with the conceptual model for enhanced polarization in the presence of ionic PFAS/HCS shown in Fig. 1.

As this experiment is an analog of a fresh AFFF contamination event, it is likely that the elevated response in σ' , as compared to the control columns, is due to an aqueous phase contaminant (i.e., PFAS or non-PFAS aqueous constituents), present in sufficiently high concentrations to increase the pore fluid conductivity. This is evidenced in part by the increase in fluid conductivity resulting from the AFFF dilution (i.e., 1.37 mS/cm to 1.43 mS/cm). This increase in electrolytic conduction may be an artifact of the experiment since

the experiment does not account for groundwater flow as would exist in the field. Flow would presumably flush and dilute aqueous phase contamination, at least in the more mobile domain of the pore space.

The NMR T_2 distributions collected on these same columns (Fig. 3) were similar over the first 10 days following sample preparation, showing little difference in the response of the experimental and control groups. Measurements on the control group after day 10 were not collected as no additional PFAS was present within the samples to alter the response. Measurements were collected on the experimental group until day 31 to confirm any subtle shifts in the T_2 distributions possibly attributable to adsorbed PFAS contamination. The T_{2ML} for all columns varied between ~ 0.02 and ~ 0.08 s, with the impacted columns exhibiting a slightly shorter T_{2ML} than the control group (~ 0.02 s versus ~ 0.03 s respectively). The T_{2ML} of the impacted group (Fig. 3b) appears to shift slightly from shorter to longer T_2 between days 0 and 24. This shift was comparable for the control group columns between days 0 and 10 (Fig. 3a). Unlike the SIP measurements, any sensitivity of NMR to fresh AFFF contamination is unclear. The synthetic samples saturated with AFFF impacted synthetic groundwater GW1 did not exhibit changes in T_2 hypothesized in Fig. 1.

4.3.2 Soils Saturated with AFFF Impacted Site Groundwater—The SIP measurements collected on synthetic soils saturated with AFFF impacted groundwater GW2 (Fig. 4) show some similarities in spectral characteristics relative to the AFFF impacted experiment (Fig. 2). Both experimental and control group columns were observed to have similar σ'' spectra to the AFFF impacted experiment columns, with a polarization peak frequency visible at ~ 80 Hz for all measurements. Samples from the experimental group (saturated with groundwater GW2) were observed to have the greatest increase in σ'' (between days 0 and 10). The control group (saturated with unimpacted synthetic groundwater based on GW2) σ'' spectra increased from ~ 0.4 mS/m to ~ 0.6 mS/m, while the experimental group σ'' spectra increased from ~ 0.5 mS/m to ~ 0.9 mS/m at their peak frequencies. σ' spectra at peak frequency increased from ~ 35 mS/m to ~ 43 mS/m for the experimental group and remained relatively unchanged at ~ 35 mS/m for the control group. Spectra were generally repeatable, resulting in similar variability as observed in the AFFF impacted experiment. Like the AFFF impacted columns, both control and impacted columns do show increases in σ'' over 10 days, but this increase was less significant than the AFFF impacted groundwater columns over a relatively similar time period (12 days). The increase in σ'' of unimpacted columns is likely due to sorption of cations present within the groundwater GW2 and its synthetic groundwater analog, rather than active sorption of ionic PFAS or HCS constituents in significant enough quantities to produce a response in σ'' .

While this appears to be in contrast with the results of the AFFF impacted experiment, it is important to note that direct sampling of groundwater from a historic AFFF impacted site will likely not result in a significant concentration of strongly sorbed AFFF/HCS constituents (e.g., strongly sorbing zwitterionic or cationic PFAS) despite high levels of contaminate concentration in the groundwater. Previous studies have shown that PFAS are not necessarily effectively desorbed via flushing with water (Schaefer et al., 2022), indicating the stability of PFAS sorbed to mineral surfaces. In contrast, during an initial

release of AFFF contamination, PFAS and HCS that associate with the mineral surface through sorption will likely be represented in much higher concentrations compared to groundwater sampled from an AFFF impacted site. As these constituents are likely responsible for the σ'' response, it is understandable why columns saturated with impacted groundwater GW2 may not produce a similar result as the columns saturated with AFFF impacted synthetic groundwater GW1. Thus, the synthetic soil experiments suggest that a sub-group of ionic PFAS and/or HCS, being those sorbed to the mineral surface, is likely responsible for observed increases in σ'' in the AFFF impacted columns (Fig. 3b).

4.2 AFFF Impacted Site Samples

Natural soils collected from the AFFF impacted site (Fig. 5) exhibit similar characteristics in the SIP data to those observed in AFFF impacted synthetic soils (Fig. 2), with the soils from the zone of known AFFF contamination having a significantly larger σ'' response (pronounced above 0.1 Hz) than the three soils outside of the known contamination zone. This difference was most significant around ~100 Hz. Within each group, however, σ'' was not well correlated to total PFAS, indicating the probable role of HCS in also contributing to the geophysical signature. This is most obvious when the σ'' spectra of sample I3 is compared to spectra from the other impacted samples. Despite having a lower total PFAS concentration than I1 and I2, I3 had a higher σ'' . The high repeatability and low standard deviations observed for the triplicate columns from both experimental and control groups build confidence in these differences being significant. In contrast, the σ'' differences are not significant between the experimental and control groups.

In a general sense, these results do match the observations from the synthetic experiments, in which exposure to AFFF results in an increase in σ'' . This result is encouraging as PFAS concentrations within source zones can be significantly higher than at the sampled PFAS impacted site (a maximum of 1340 $\mu\text{g/kg}$, Table S1). As an example, soil PFAS concentrations up to order of magnitude higher (i.e. 70,000 $\mu\text{g/kg}$ (McGuire et al., 2014)) have been recorded at another AFFF source zone.

The T_2 distributions from the NMR measurements also show some variation between sites that are not supported by differences in the textural characteristics (Fig. 6). Samples from the experimental group had T_2 distributions with shorter relaxation time components, and T_{2ML} values centered at shorter relaxation times. However, the T_{2ML} values calculated from the T_{2ML} distributions were comparable for the two groups. Thus, these results do not corroborate the results from the synthetic experiment. While there is some differentiation between T_{2ML} of the impacted and control soil groups, the impacted soils appear to have shorter T_{2ML} , rather than longer T_{2ML} as was observed for the synthetic data. The NMR data from both the synthetic and AFFF impacted soil experiments, therefore do not suggest any relationship between AFFF exposure and NMR response.

4.3 Potential mechanisms of complex resistivity sensitivity

The mechanism described in Fig. 1 represents the basic conceptual model for how AFFF contamination might result in an IP response: ionic constituents of AFFF sorb to the surface,

resulting in an increase in surface ionic strength and a subsequent increase in charge polarization. This mechanism may be overly simplistic in the case of PFAS contamination as it overlooks other consequential factors, discussed briefly here.

Organic Content of Soils —Organic content, characterized as TOC, has been shown to be an important factor for PFAS association with the soil (Nguyen et al., 2020). PFAS soil concentrations have been observed to be higher in areas with elevated TOC. Variations in TOC, may result in a similar polarization response (Yu et al., 2014). Consequently, the response from AFFF contamination and variations in TOC may be non-unique. TOC from the field site in this study varied minimally, lending evidence that the response exhibited in Fig. 5 was likely not due to TOC variability, but rather a result of ionic AFFF constituents (likely cationic or zwitterionic PFAS constituents or non-PFAS ionic constituents such as ionic HCS).

In addition to non-uniqueness between AFFF contamination and TOC variations, another factor is the complexity of the polarization associated with sorption onto organic soils. The complex resistivity response to sorption onto organic matter is currently poorly understood.

Effect of Non-Ionic HCS —Another factor complicating our understanding of how AFFF contamination may influence the CR response is the potential influence of non-ionic HCS found in AFFF. Surfactants have been shown to have a buffering effect on polarization, in which non-ionic surfactants might broaden the dual double layer, or act as a secondary electrical double layer (Mellage et al., 2018). SIP measurements might consequently exhibit changes in the peak frequency of the observed spectra associated with this phenomenon. Previous experiments have observed a negative correlation between σ'' and non-ionic HCS concentration (Mellage et al., 2018) or no effect of non-ionic HCS on polarization (Deng et al., 2020). The effect observed by Mellage et al. (2018) is likely not the dominant sensitivity mechanism influencing the samples in this study given that a positive correlation between σ'' and AFFF exposure is observed (Fig. 2, 4, and 5). However, the presence of non-ionic HCS in AFFF, and potential variations in these additives dependent on different AFFF products, could complicate interpretation of complex resistivity data due to competing polarization mechanisms.

Anionic vs. cationic/zwitterionic AFFF constituents —Our conceptual model primarily highlights the potential effect of sorption of cationic/zwitterionic AFFF constituents onto net-negatively charged mineral surfaces. Anionic AFFF constituents, however, *might also impact SIP measurements*. The increased concentration of cationic/zwitterionic cations at the surface, combined with an increase in the concentration of anions in solution, could produce complex charge dynamics within the electrical double layer. Similarly to the phenomena discussed in regards to HCS, varying anions could result in competing polarization mechanisms.

4.4 Implications for source zone characterization

The results from the experiments conducted in this study present an early evaluation of the likely sensitivity of SIP and NMR measurements to AFFF contamination across

source zones. Although NMR T_{2ML} measurements were not sufficiently sensitive to AFFF contamination, these results provide encouraging first evidence for a case to use SIP to characterize AFFF contamination within a source zone. However, the column experiments presented in this study were conducted under near-ideal circumstances, using relatively homogeneous soils with well-controlled saturating fluids. It is well known that σ'' is sensitive to physical variations in soils in addition to interfacial chemical processes, such as sorption dynamics. Variations in soil physical characteristics might obscure any spatial variations in sorbed-PFAS concentrations across a source zone. This uncertainty is often an issue for geophysical characterization; combining SIP with other methods sensitive to soil characteristics, yet insensitive to PFAS contamination, might limit this problem.

4.5 Future research

In order to effectively use CR for PFAS source zone characterization, future research should focus on better understanding the mechanisms of polarization exhibited by AFFF contaminated soils. This research should focus on more controlled experiments to narrow down the individual contributions of organic material, HCS type and concentration, as well as the effect of individual AFFF constituents. Follow up research could include experiments to isolate specific AFFF constituents (for example PFAS versus HCS, cationic versus zwitterionic constituents, etc.) that contribute most to the SIP signal. Efforts to control uncertainty caused by non-uniqueness associated with varying soil properties and variations in the degree of AFFF contamination are also necessary for more accurate survey interpretation.

5. Conclusions

Despite uncertainty in the exact mechanism of CR sensitivity to AFFF contamination and typical issues of data interpretation resulting from natural variations in soil characteristics, the likely sensitivity of the SIP method to both sorbed PFAS and/or sorbed HCS constituents may ultimately facilitate efficient, cost-effective geophysical characterization of AFFF source zone contamination. While it is too early to presume that SIP will be effective in meeting this need, findings from this study warrant further research to better define the sensitivity of SIP to soil AFFF contamination and the mechanisms underlying this response. However, we emphasize again that the primary value of these geophysical measurements will be field-scale deployment for efficient characterization of variations in AFFF contamination across source zones, leading to a more informed direct soil sampling strategy that would optimize the value of subsequent PFAS analysis.

Supplementary Material

Refer to Web version on PubMed Central for supplementary material.

Acknowledgments

This material is based upon work supported by the Humphreys Engineer Center Support Activity under Contract No. W912HQ19P0043. Work was conducted under Strategic Environmental Research and Development Program (SERDP) project #ER19-1128. Views, opinions, and/or findings contained in this report are those of the author and should not be construed as an official department of defense position or decision unless so designated by other

official documentation. We thank Dung Nguyen and Sean O'Hare of CDM Smith for their contributions to ongoing laboratory and field assistance. We also thank Katherine Rodriguez and Carolina Caro for laboratory support in the collection of some of the data presented in this study.

The views expressed in this article are those of the authors and do not necessarily represent the views or policies of the U.S. Environmental Protection Agency. Mention of trade names or commercial products does not constitute endorsement or recommendation for use.

References

- Anderson RH, Long GC, Porter RC, Anderson JK, 2016. Occurrence of select perfluoroalkyl substances at U.S. Air Force aqueous film-forming foam release sites other than fire-training areas: Field-validation of critical fate and transport properties. *Chemosphere* 150, 678–685. 10.1016/j.chemosphere.2016.01.014 [PubMed: 26786021]
- Atekwana E, Slater LD, 2009. Biogeophysics : a New Frontier in Earth Science Research. *Rev. Geophys.* 47, 1–30. 10.1029/2009RG000285
- Backe WJ, Day TC, Field JA, 2013. Zwitterionic, Cationic, and Anionic Fluorinated Chemicals in Aqueous Film Forming Foam Formulations and Groundwater from U.S. Military Bases by Nonaqueous Large-Volume Injection HPLC-MS/MS. *Environ. Sci. Technol.* 47, 5226–5234. 10.1021/es3034999 [PubMed: 23590254]
- Barzen-Hanson KA, Roberts SC, Choyke S, Oetjen K, McAlees A, Riddell N, McCrindle R, Ferguson PL, Higgins CP, Field JA, 2017. Discovery of 40 Classes of Per- and Polyfluoroalkyl Substances in Historical Aqueous Film-Forming Foams (AFFFs) and AFFF-Impacted Groundwater. *Environ. Sci. Technol.* 51, 2047–2057. 10.1021/acs.est.6b05843 [PubMed: 28098989]
- Ben Moshe S, Furman A, 2022. Real-time monitoring of organic contaminant adsorption in activated carbon filters using spectral induced polarization. *Water Res.* 212, 118103. 10.1016/j.watres.2022.118103 [PubMed: 35081496]
- Brusseau ML, Yan N, Van Glubt S, Wang Y, Chen W, Lyu Y, Dungan B, Carroll KC, Holguin FO, 2019. Comprehensive retention model for PFAS transport in subsurface systems. *Water Res.* 148, 41–50. 10.1016/j.watres.2018.10.035 [PubMed: 30343197]
- Bryar TR, Knight RJ, 2008. NMR relaxation measurements to quantify immiscible organic contaminants in sediments. *Water Resour. Res.* 44. 10.1029/2006WR005635
- Campos Pereira H, Ullberg M, Kleja DB, Gustafsson JP, Ahrens L, 2018. Sorption of perfluoroalkyl substances (PFASs) to an organic soil horizon – Effect of cation composition and pH. *Chemosphere* 207, 183–191. 10.1016/j.chemosphere.2018.05.012 [PubMed: 29793030]
- Day-Lewis FD, Slater LD, Robinson J, Johnson CD, Terry N and Werkema D, 2017. An overview of geophysical technologies appropriate for characterization and monitoring at fractured-rock sites. *Journal of environmental management.* 204. 709–720. [PubMed: 28434821]
- Death C, Bell C, Champness D, Milne C, Reichman S and Hagen T, 2021. Per-and polyfluoroalkyl substances (PFAS) in livestock and game species: A review. *Sci. Total Environ.* 774, 144795 [PubMed: 33609849]
- Deng Y, Shi X, Zhang Z, Sun Y, Wu J, Qian J, 2020. Application of spectral induced polarization for characterizing surfactant-enhanced DNAPL remediation in laboratory column experiments. *J. Contam. Hydrol.* 230, 103603. 10.1016/j.jconhyd.2020.103603 [PubMed: 31980237]
- García RA, Chiaia-Hernández AC, Lara-Martin PA, Loos M, Hollender J, Oetjen K, Higgins CP, Field JA, 2019. Suspect Screening of Hydrocarbon Surfactants in AFFFs and AFFF-Contaminated Groundwater by High-Resolution Mass Spectrometry. *Environ. Sci. Technol.* 53, 8068–8077. 10.1021/acs.est.9b01895 [PubMed: 31269393]
- Hao N, Moysey SMJ, Powell BA, Ntarlagiannis D, 2016. Comparison of the surface ion density of silica gel evaluated via spectral induced polarization versus acid-base titration. *J. Appl. Geophys.* 135, 427–435. 10.1016/j.jappgeo.2016.01.014
- Hao N, Moysey SMJ, Powell BA, Ntarlagiannis D, 2015. Evaluation of Surface Sorption Processes Using Spectral Induced Polarization and a 22 Na Tracer. *Environ. Sci. Technol.* 49, 9866–9873. 10.1021/acs.est.5b01327 [PubMed: 26191613]

- Houtz EF, Higgins CP, Field JA, Sedlak DL, 2013. Persistence of Perfluoroalkyl Acid Precursors in AFFF-Impacted Groundwater and Soil. *Environ. Sci. Technol.* 47, 8187–8195. 10.1021/es4018877 [PubMed: 23886337]
- Keating K, Knight R, 2007. A laboratory study to determine the effect of iron oxides on proton NMR measurements. *Geophysics* 72, E27. 10.1190/1.2399445
- Kessouri P, Furman A, Huisman JA, Martin T, Mella A, Ntarlagiannis D, Bückner M, Ehosioke S, Fernandez P, Flores-Orozco A, others, 2019. Induced polarization applied to biogeophysics: recent advances and future prospects. *Near Surf. Geophys.* 17, 595–621.
- Kirmizakis P, Kalderis D, Ntarlagiannis D, Soupios P, 2019. Preliminary assessment on the application of biochar and spectral induced polarization for wastewater treatment. *Near Surf. Geophys.* 1–14. 10.1002/nsg.12076
- Kleinberg RL, Horsfield MA, 1990. Transverse relaxation processes in porous sedimentary rock. *J. Magn. Reson.* 88, 9–19. 10.1016/0022-2364(90)90104-H
- McGuire ME, Schaefer C, Richards T, Backe WJ, Field JA, Houtz E, Sedlak DL, Guelfo JL, Wunsch A, Higgins CP, 2014. Evidence of Remediation-Induced Alteration of Subsurface Poly- and Perfluoroalkyl Substance Distribution at a Former Fire fighter Training Area.
- McGuire ME, Schaefer C, Richards T, Backe WJ, Field JA, Houtz E, Sedlak DL, Guelfo JL, Wunsch A, Higgins CP, 2014. Evidence of remediation-induced alteration of subsurface poly- and perfluoroalkyl substance distribution at a former firefighter training area. *Environ. Sci. & Technol.* 48, 6644–6652. [PubMed: 24866261]
- Mella A, Holmes AB, Linley S, Vallée L, Rezanezhad F, Thomson N, Gu FX, Van Cappellen P, 2018. Sensing coated iron-oxide nanoparticles with spectral induced polarization (SIP): Experiments in natural sand packed flow-through columns. *Environ. Sci. Technol.* acs.est.8b03686. 10.1021/acs.est.8b03686
- Mella A, Zakai G, Efrati B, Pagel H, Schwartz N, 2022. Paraquat sorption-and organic matter-induced modifications of soil spectral induced polarization (SIP) signals. *Geophys. J. Int.* 229, 1422–1433.
- Nguyen TMH, Bräunig J, Thompson K, Thompson J, Kabiri S, Navarro DA, Kookana RS, Grimison C, Barnes CM, Higgins CP, others, 2020. Influences of chemical properties, soil properties, and solution pH on soil--water partitioning coefficients of per- and polyfluoroalkyl substances (PFASs). *Environ. Sci. & Technol.* 54, 15883–15892. [PubMed: 33249833]
- Schaefer C, Dung N, Emerson C, Stefanie S, P. HC, Jennifer F, 2022. Desorption Isotherms for Poly- and Perfluoroalkyl Substances in Soil Collected from an Aqueous Film-Forming Foam Source Area. *J. Environ. Eng.* 148, 4021074. 10.1061/(ASCE)EE.1943-7870.0001952
- Schaefer CE, Choyke S, Ferguson PL, Andaya C, Burant A, Maizel A, Strathmann TJ, Higgins CP, 2018. Electrochemical Transformations of Perfluoroalkyl Acid (PFAA) Precursors and PFAAs in Groundwater Impacted with Aqueous Film Forming Foams. *Environ. Sci. Technol.* 52, 10689–10697. 10.1021/acs.est.8b02726 [PubMed: 30130962]
- Tang CY, Shiang Fu Q, Gao D, Criddle CS, Leckie JO, 2010. Effect of solution chemistry on the adsorption of perfluorooctane sulfonate onto mineral surfaces. *Water Res.* 44, 2654–2662. 10.1016/j.watres.2010.01.038 [PubMed: 20172580]
- Taylor MD, 2020. Animal size impacts perfluoroalkyl acid (PFAA) concentrations in muscle tissue of estuarine fish and invertebrate species. *Environ. Pol.* 267, 115595. 10.1016/j.envpol.2020.115595.
- US Environmental Protection Agency (USEPA), 2022. EPA PFAS Drinking Water Laboratory Methods [WWW Document]. URL <https://www.epa.gov/pfas/epa-pfas-drinking-water-laboratory-methods>
- Vaudelet P, Revil A, Schmutz M, Franceschi M, Bégassat P, 2011. Induced polarization signatures of cations exhibiting differential sorption behaviors in saturated sands. *Water Resour. Res.* 47, 1–21. 10.1029/2010WR009310
- Weller A, Slater L, 2019. Permeability estimation from induced polarization: an evaluation of geophysical length scales using an effective hydraulic radius concept. *Near Surf. Geophys.* 1–14. 10.1002/nsg.12071

- Yu G, Hu W, He Z, Xiang K, Hu H, He L, Wang T, Li P, 2014. Complex Resistivity Characteristics of High TOC Marine Shale Core Samples and its Applications, in: SEG International Exposition and Annual Meeting. p. SEG-2014-0512.
- Zhang C, Slater L, Redden G, Fujita Y, Johnson T, Fox D, 2012. Spectral induced polarization signatures of hydroxide adsorption and mineral precipitation in porous media. Environ. Sci. Technol. 46, 4357–4364. 10.1021/es204404e [PubMed: 22420512]

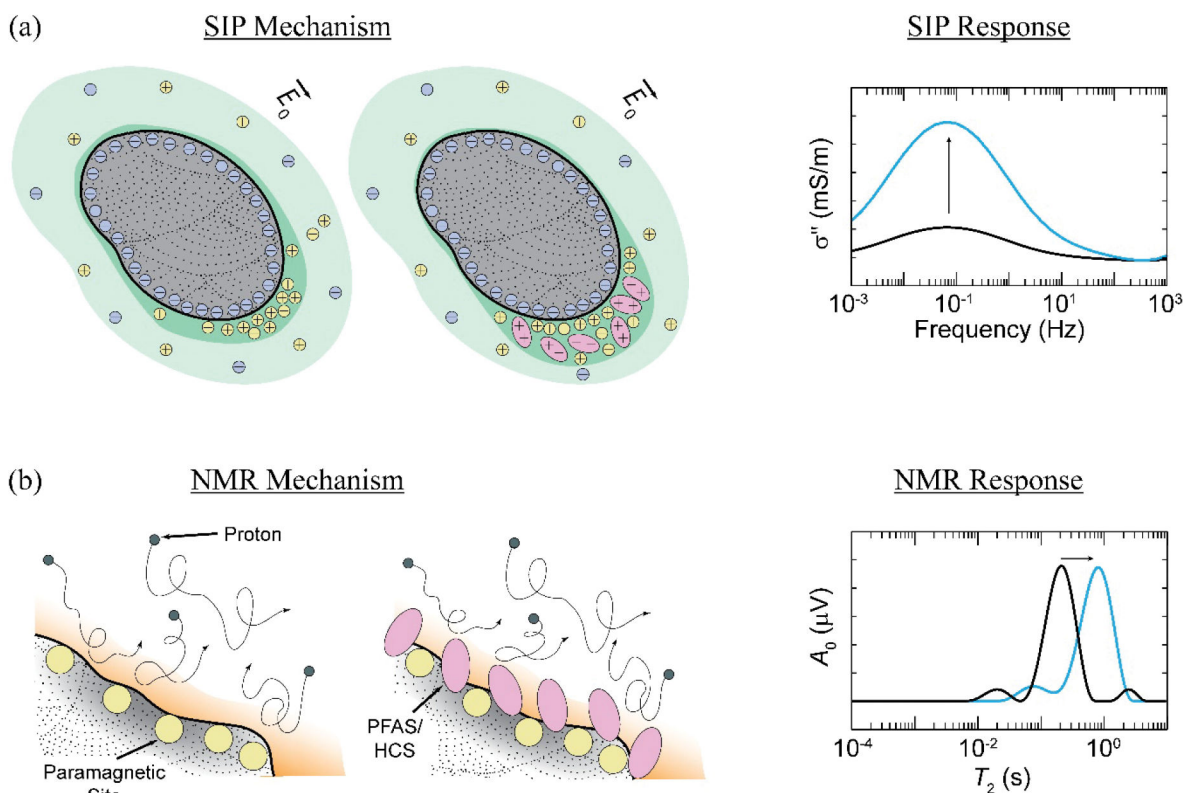


Figure 1.

A conceptual illustration of the hypothetical mechanisms and associated responses (blue) of (a) SIP and (b) NMR to sorbed ionic PFAS/HCS as compared to baseline (black). SIP is theorized to exhibit augmented polarization, resulting in a measurable increase in σ'' , while NMR is theorized to exhibit diminished proton/paramagnetic coupling at the surface resulting in a measurable increase in T_2 . The blue and black lines in the SIP and NMR responses are hypothetical measurements of samples with and without PFAS, respectively.

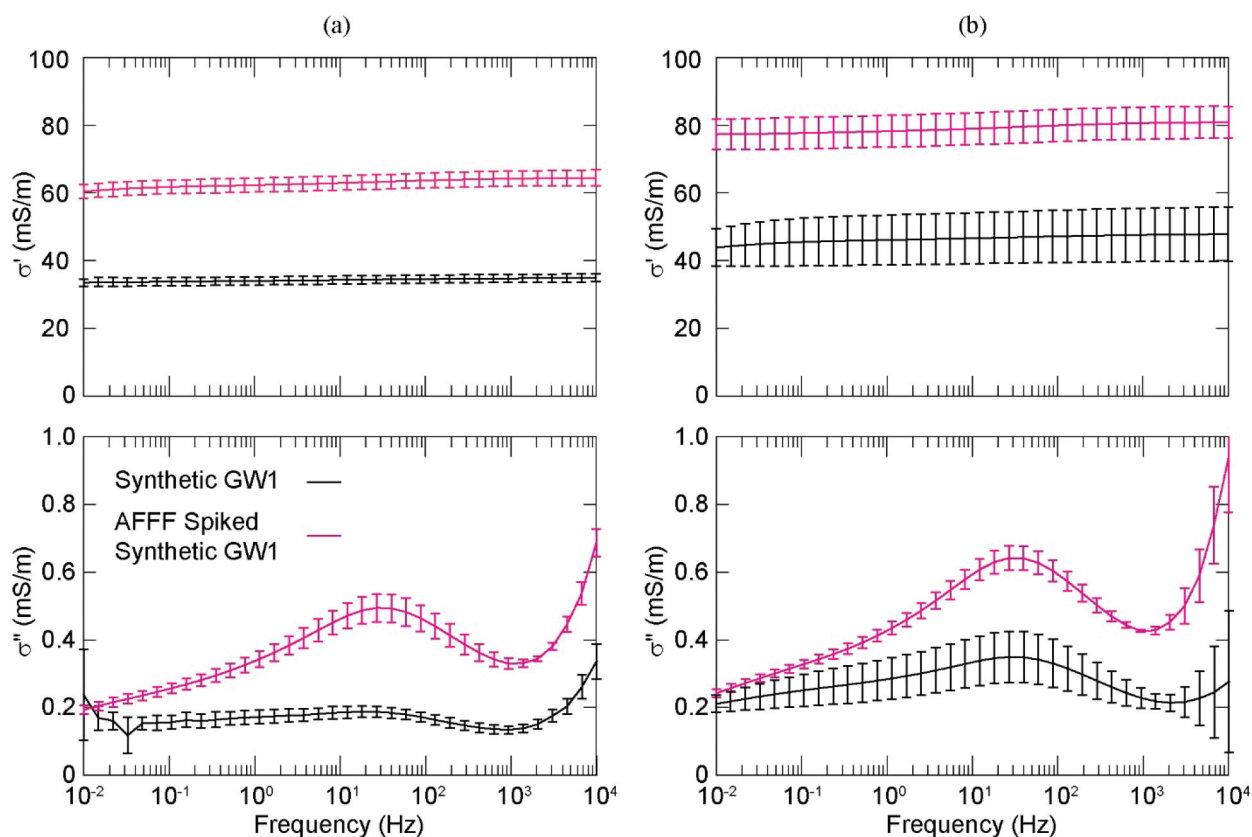


Figure 2. Comparison of the SIP measurements collected on columns packed with synthetic soil, saturated with clean (black) and AFFF impacted synthetic groundwater (pink), both based on sampled groundwater GW1. The σ' and σ'' spectra from days 0 and 12 (a and b respectively). Error bars are the standard deviation from columns prepared in triplicate.

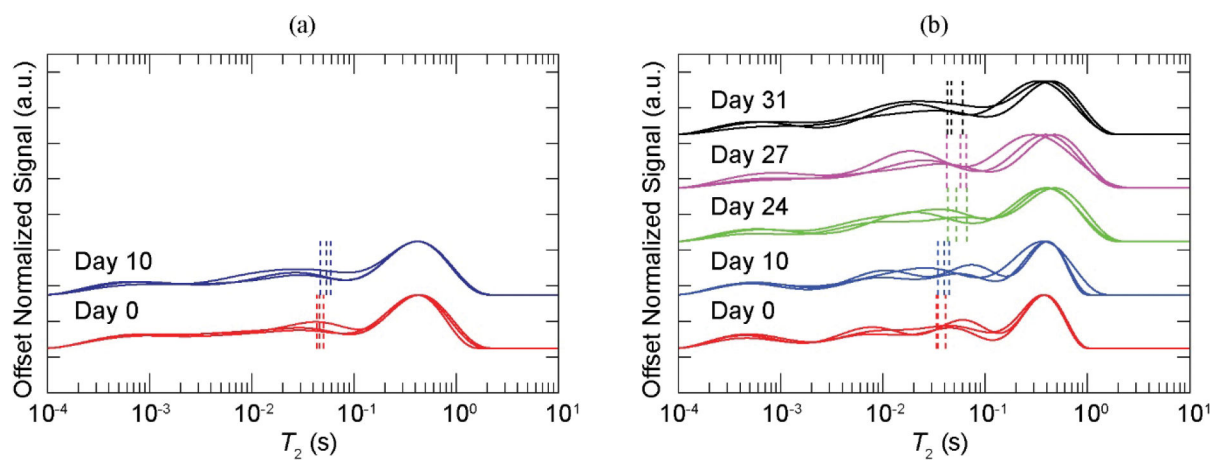


Figure 3. NMR measurements collected from the synthetic samples saturated with (a) synthetic groundwater based on GW1 and (b) AFFF impacted synthetic groundwater based on GW1. The T_2 distributions are plotted in triplicate, with vertical lines representing T_{2ML} of each distribution.

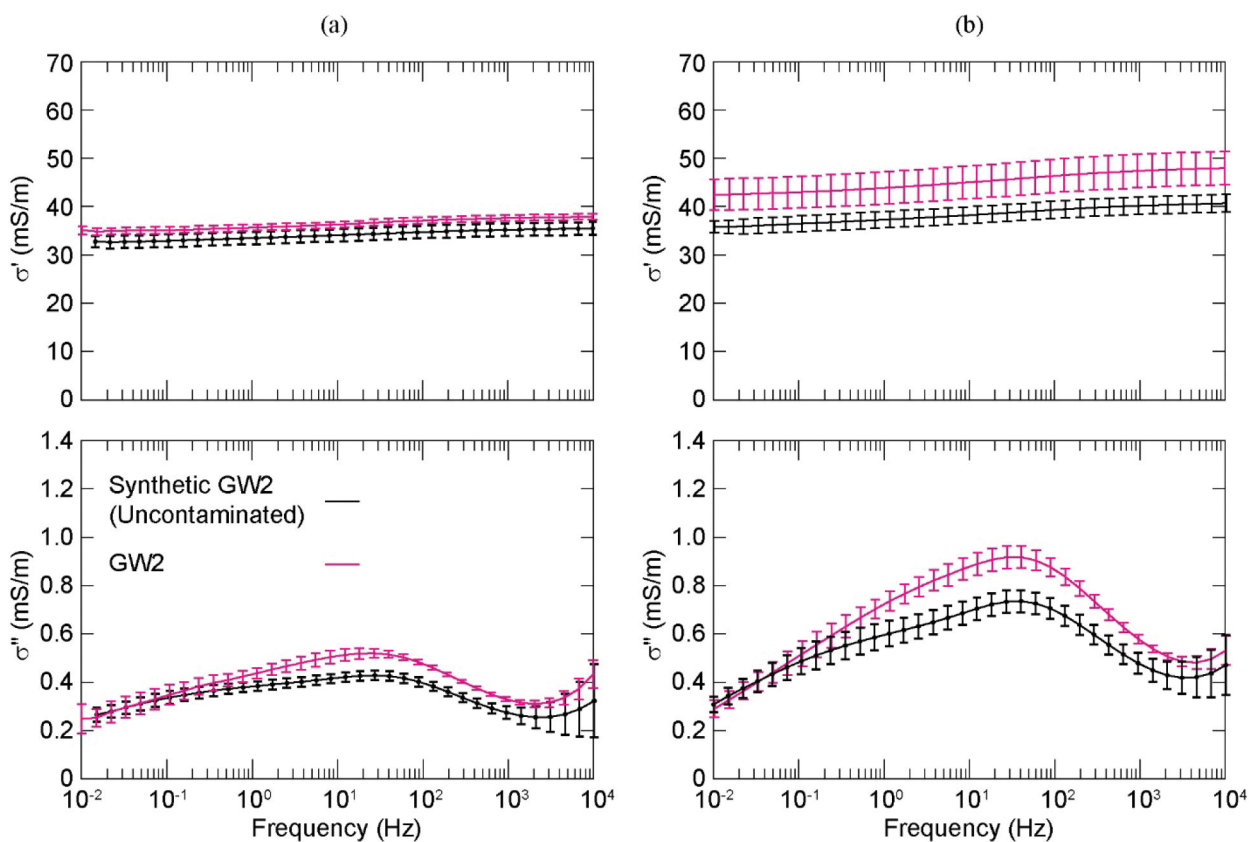


Figure 4.

A comparison of the SIP measurements collected on columns packed with synthetic soil, saturated with clean synthetic groundwater (black) based on GW2, and AFFF impacted groundwater (pink) GW2. The σ' and σ'' spectra from days 0 and 10 (a and b respectively). Error bars are the standard deviation from columns prepared in triplicate.

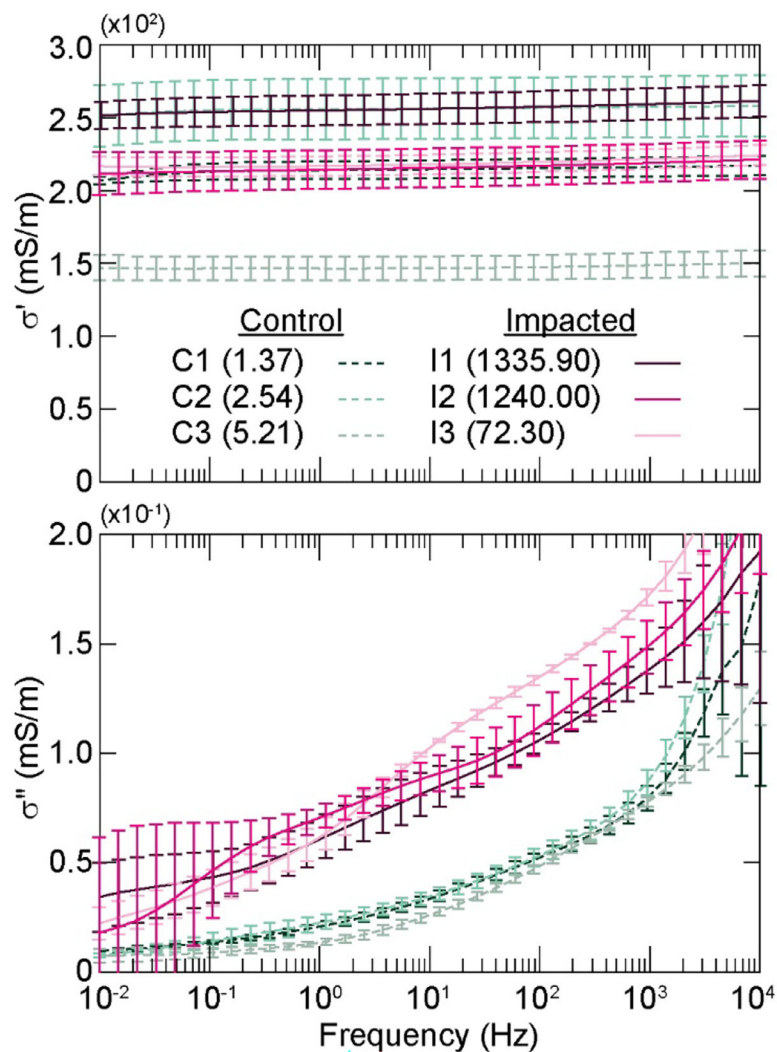


Figure 5.

The SIP data, including σ' and σ'' , collected on the soils obtained from the AFFF impacted site. Control group samples, collected from the unimpacted region of the site are labeled C1-C3, and the experimental group samples, collected from the source zone of contamination, are labeled I1-I3. Error bars are standard deviations calculated from columns prepared in triplicate. Total PFAS concentrations ($\mu\text{g/kg}$) are included in parentheses.

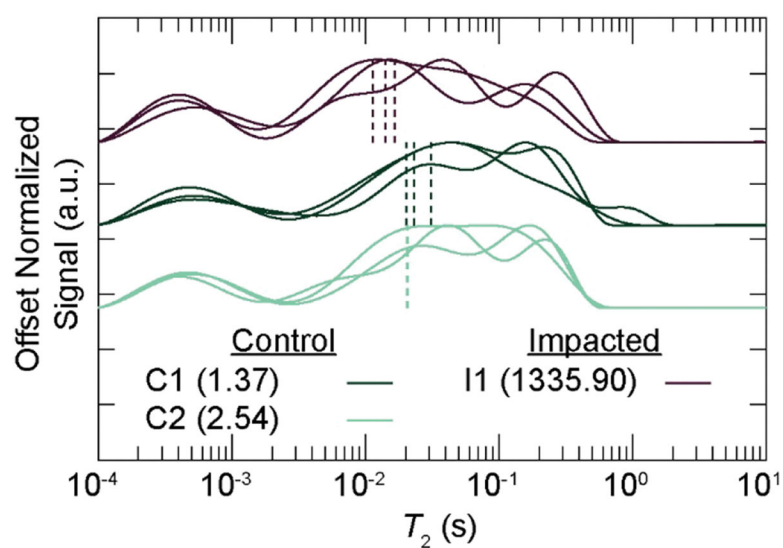


Figure 6. Examples of NMR data collected on selected soil samples from the AFFF impacted site. The T_2 distributions are plotted in triplicate, with vertical lines representing T_{2ML} of each distribution.